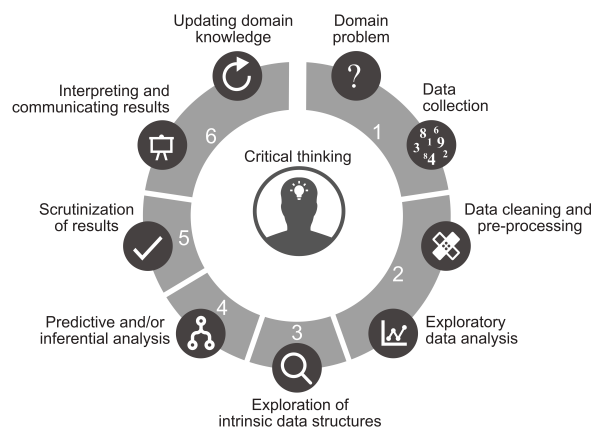


| STAT214 Chapter 2 The Data Science Life Cycle

| 1 Data Terminology

| 1.1 The Data Science Life Cycle

1. Domain problem formulation and data collection
2. Data cleaning, preprocessing, and exploratory data analysis
3. Exploration of intrinsic data structures (dimensionality reduction and clustering)
4. Predictive and/or inferential analysis
5. Evaluation of results
6. Communication of results



| 1.2 Variable

Each **column** of a data table corresponds to a different type of measurement and these are called the **variables**, **features**, **attributes**, or **covariates** of the data. Each variable in a dataset typically has one of the following types:

- Numeric
- Categorical
- Dates and times
- Structured (short) text
- Unstructured (long) text

| 1.3 Dimension

The **dimension** of the data refers to the number of variables (columns) that it contains

| 1.4 Observation

Each row corresponds to a particular **observation**, **observational unit**, **data unit**, or **data point**.

| 2 DSLC Stage 1: Problem Formulation and Data Collection

| 2.1 Formulating a Question

- Collaborate with domain experts to reformulate a domain question into one that can be answered using available data.
- After nailing down the goals of the project, the next task is to identify what data is already available or can be collected to answer it.

| 2.2 Data Collection

- While some projects will use existing data, other projects will involve collecting new data.
- When using existing data that has already been collected, ideally, you will be working with **live data** that is kept up to date.
- Data that was collected in the past and is now stagnant, and for which no additional relevant information can be obtained, is considered **dead data**.

⚠ Remark: Suggestion ▾

Real data almost never perfectly reflects reality, but knowing how a dataset was collected can give some sense of **how** the data is related to reality. At **minimum**, every dataset that you use should be accompanied by detailed data documentation (such as **README** files and codebooks) that describes how the data was collected and what measurements it contains.

| 2.3 Making a Plan for Evaluating Predictability

Before moving to the second stage of the DSLC, we recommend making a plan for evaluating the predictability of your eventual results.

| 3 DSLC Stage 2: Data Cleaning and Exploratory Data Analysis

| 3.1 Data Cleaning

- Identify issues with the data (such as strange formatting and invalid values)
- modify it so that its values are valid and it is formatted in a computational- and human-friendly way
- helps you to develop a more detailed understanding of the information that the data contains as well as its limitations
- is necessarily subjective and involves making assumptions about the underlying real-world quantities being measured

| 3.2 Preprocessing

Preprocessing refers to the process of modifying your cleaned data to **fit the requirements of a specific algorithm** that you want to apply:

- Scale transformation
- Impute or remove missing values
- Featurization

| 3.3 Exploratory Data Analysis

- **exploratory data analysis (EDA)**: involves developing rough numeric and visual summaries of the data that will help you understand the data and the patterns that it contains
- **explanatory data analysis**: involves polishing the most informative exploratory tables and graphs to communicate them to external audiences

| 3.4 Data Snooping and PCS

Data snooping:

A phenomenon in which you **present the relationships and patterns that you found while exploring your data as proven conclusions** when in reality, all you have done is show that these patterns exist for one specific set of data points and sequence of judgment calls

Avoid data snooping:

- analyses be preregistered before they are undertaken (may prevent the discovery of new patterns and relationships)
- demonstrating the predictability and stability of your results

| 4 DSLC Stage 3: Uncovering Intrinsic Data Structures

Dig a little bit deeper by teasing out some more complex relationships and patterns:

- Dimensionality reduction analysis
 - Cluster analysis
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| 5 DSLC Stage 4: Predictive and/or Inferential Analysis

| 5.1 Prediction

Use past or current observable data to predict something about future unseen data, usually to help make real-world decisions

- **supervised learning algorithms**: classification and regression algorithms
- **unsupervised learning algorithms**: clustering and dimensionality analysis

| 5.2 Data-Driven Inference

Learning about a broader population by quantifying the uncertainty associated with a **parameter estimate**

| 6 DSLC Stage 5: Evaluation of Results

- should be performed throughout the DSLC
 - use critical thinking and quantitatively evaluating your results using the PCS framework
 - recommend using a **negative control**
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| 7 DSLC Stage 6: Communication of Results and Updating Domain Knowledge

Communicate your results to your intended audience so they can be used for real-world decision making.

- communicate data and results in the context of creating impactful data visualizations
- should be tailored to your audience
- put the results “into production”